Problem Set 1 Solutions

AI 501: Mathematics for Artificial Intelligence

Fall 2026

Problem 1. Solution. $y = 2x - \mathbf{1}$ (each 0 maps to -1, each 1 maps to $+1$). Inverting, $x = \frac{1}{2}(y + \mathbf{1})$.

Problem 2. Solution. (a) $\mathbf{1}^T s$. (b) $(e_1 + e_2 + \cdots + e_{10})^T s = 5$, i.e. $(\mathbf{1}_{10}, \mathbf{0}_{10})^T s = 5$, where $(\mathbf{1}_{10}, \mathbf{0}_{10})$ is the 20-vector formed by stacking ten ones on top of ten zeros.

Problem 3. Solution. (a) $\mathbf{1}^T w$ is the total number of words in the document. (b) It means dictionary word #282 never appears in the document. (c) $h = w/(\mathbf{1}^T w)$.

Problem 4. Solution. Introduce the 5-vector ρ of exchange rates, where ρ_i is the value of one unit of currency i in USD ($\rho_1 = 1$ for USD itself). Total value in USD $= \rho^T c$.

Problem 5. Solution. Here $w = (0, 2, -1)$ is read as a 1×3 row vector and $z = (-3, 2)$ as a 1×2 row vector, so w^T is 3×1 and z^T is 2×1. Recall that for a matrix product AB to be defined, the number of columns of A must equal the number of rows of B.

(a) $w + z$: **invalid** — addition requires matching shapes, and w is 1×3 while z is 1×2.

(b) $w^T z$: **valid.** w^T is 3×1 and z is 1×2; the inner dimension is $1 = 1$, so the product is defined and gives a 3×2 matrix (the outer product of w and z), with entries $(w^T z)_{ij} = w_i z_j$:

$$w^T z = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} \begin{bmatrix} -3 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -6 & 4 \\ 3 & -2 \end{bmatrix}.$$

(c) $z^T w$: **valid**, by the same reasoning: z^T is 2×1, w is 1×3, inner dimension $1 = 1$, giving a 2×3 matrix:

$$z^T w = \begin{bmatrix} -3 \\ 2 \end{bmatrix} \begin{bmatrix} 0 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 0 & -6 & 3 \\ 0 & 4 & -2 \end{bmatrix}.$$

Note $z^T w = (w^T z)^T$, consistent with the identity $(AB)^T = B^T A^T$ — you can see this by comparing the two matrices above.

(d) wz: **invalid.** w is 1×3 and z is 1×2; the inner dimension would need columns of w (3) to equal rows of z (1), and $3 \neq 1$.

So $w^T z$ and $z^T w$ *are* valid — each pairs a $\cdot \times 1$ matrix with a $1 \times \cdot$ matrix, which always multiplies — while $w + z$ and wz fail because their shapes never line up.

Problem 6. Solution. Write $b_1 = \alpha_1 a_1 + \alpha_2 a_2$ and $b_2 = \beta_1 a_1 + \beta_2 a_2$, and $c = \gamma_1 b_1 + \gamma_2 b_2$. Substituting,

$$c = \gamma_1(\alpha_1 a_1 + \alpha_2 a_2) + \gamma_2(\beta_1 a_1 + \beta_2 a_2) = (\gamma_1\alpha_1 + \gamma_2\beta_1)a_1 + (\gamma_1\alpha_2 + \gamma_2\beta_2)a_2,$$

which is a linear combination of a_1, a_2.

Problem 7. Solution. (a) Tomorrow's predicted price is just today's price (a "persistence" forecast). (b) Tomorrow's predicted price extrapolates the trend from yesterday to today: predicted price = 2×today's price − yesterday's price. (c) Tomorrow's predicted price is simply the price recorded 5 days ago, i.e. z_{t-5} (six days before the day being predicted). (d) Tomorrow's predicted price is the average of today's and yesterday's prices.

Problem 8. Solution. Each component must be rescaled so that a perfect score in it contributes its full weight out of 100. Homework: the eight grades total at most 80, contributing 25 points, so each homework entry has weight 25/80. Midterm: 35/120. Final: 40/160. Hence

$$w = (0.3125, 0.3125, 0.3125, 0.3125, 0.3125, 0.3125, 0.3125, 0.3125,\ 0.2917,\ 0.2500),$$

i.e. $w_i = 25/80 = 0.3125$ for $i = 1, \ldots, 8$, $w_9 = 35/120 = 0.2917$, $w_{10} = 40/160 = 0.2500$. Check: a perfect record gives $80(0.3125) + 120(35/120) + 160(0.25) = 25 + 35 + 40 = 100$.

Problem 9. Solution. (a) A general linear combination (no restriction) traces out the span of x_1, x_2: all of $\mathbb{R}^2$ if x_1, x_2 are not parallel, and only the line through the origin along their common direction if they are. (b) A conic combination ($\alpha_i \geq 0$) traces out the cone (infinite wedge) between the directions of x_1 and x_2 (degenerating to a ray or a line through the origin if x_1, x_2 are parallel). (c) An affine combination ($\alpha_1 + \alpha_2 = 1$) traces out the entire line through x_1 and x_2 (assuming $x_1 \neq x_2$). (d) A convex combination ($\alpha_i \geq 0$, $\alpha_1 + \alpha_2 = 1$) traces out exactly the line segment joining x_1 and x_2.

Problem 10. Solution. Substituting $\alpha_3 = 1 - \alpha_1 - \alpha_2$ gives $y = x_3 + \alpha_1(x_1 - x_3) + \alpha_2(x_2 - x_3)$, a linear combination (with free coefficients α_1, α_2) of the two vectors $x_1 - x_3$, $x_2 - x_3$, shifted by x_3. If x_1, x_2, x_3 are not collinear, this affine hull is the entire 2-dimensional plane through the three points.

Problem 11. Solution. If v, w lie on the same line through the origin (one is a scalar multiple of the other), every combination $cv + dw$ also lies on that line, so the span is just that line. Otherwise, the combinations $cv + dw$ fill the entire plane through the origin containing v and w.

Problem 12. Solution. The equation $cu + dv + ew = b$ gives the system

$$2c - d = 1, \qquad -c + 2d - e = 0, \qquad -d + 2e = 0.$$

From the third equation $d = 2e$; substituting into the second gives $c = 3e$; substituting into the first gives $6e - 2e = 1 \Rightarrow 4e = 1 \Rightarrow e = \frac{1}{4}$. Hence $c = \frac{3}{4},\ d = \frac{1}{2},\ e = \frac{1}{4}$, and indeed b lies in the span of $\{u, v, w\}$.

Problem 13. Solution. Every one of v_1, v_2, v_3, v_4 has fifth entry equal to 0, so any linear combination $av_1 + bv_2 + cv_3 + dv_4$ also has fifth entry 0. But the target vector $(8, 5, -2, 3, 1)$ has fifth entry $1 \neq 0$. Hence the system is inconsistent and $(8, 5, -2, 3, 1)$ is **not** in the span of $\{v_1, v_2, v_3, v_4\}$ — this follows immediately by inspecting the last coordinate, with no elimination required.

Problem 14. Solution. Because a, b are orthogonal ($a^T b = 0$), the cross term vanishes when expanding the squared norm:

$$\|\alpha a + \beta b\|^2 = \alpha^2 \|a\|^2 + 2\alpha\beta\, a^T b + \beta^2 \|b\|^2 = \alpha^2 \|a\|^2 + \beta^2 \|b\|^2,$$

so $\|\alpha a + \beta b\| = \sqrt{\alpha^2\|a\|^2 + \beta^2\|b\|^2}$.

Problem 15. Solution. (c) it makes no sense. x^2 would mean the product xx of an $n \times 1$ matrix with an $n \times 1$ matrix, which is undefined for $n > 1$ (the column count 1 does not match the row count n). The inner product of x with itself is written with an explicit transpose, $x^T x$, and the vector of squared entries would need its own notation; neither is what x^2 denotes.

Problem 16. Solution. (a) **(ii) sometimes.** $\|x\| \geq 0$ always, while $-\| - x\| = -\|x\| \leq 0$ (using $\| - x\| = \|x\|$); the two sides can only agree when both equal 0, i.e. $x = 0$. (b) **(iii)** $\sqrt{2}\,\|x\|$, since $\|(x, x)\| = \sqrt{\|x\|^2 + \|x\|^2} = \sqrt{2}\,\|x\|$.